

```

1  =====
2  Generated by:      Genus(TM) Synthesis Solution 18.14-s037_1
3  Generated on:      Nov 18 2025  08:52:38 pm
4  Module:           mac
5  Operating conditions:  typical
6  Interconnect mode:  global
7  Area mode:        physical library
8  =====
9
10 Instance Module   Cell Count   Cell Area   Net Area   Total Area
11 -----
12 mac               124         348.460    168.121    516.581

```

```

=====
1   Generated by:      Genus(TM) Synthesis Solution 18.14-s037_1
2   Generated on:      Nov 18 2025  08:52:26 pm
3   Module:           mac
4   Operating conditions: typical
5   Interconnect mode: global
6   Area mode:        physical library
7   =====
8
9
10
11 Path 1: MET (43 ps) Setup Check with Pin mult_reg_reg[6]/CK->D
12 |    Group: clk
13 | Startpoint: (R) W[1]
14 |    Clock: (R) clk
15 | Endpoint: (R) mult_reg_reg[6]/D
16 |    Clock: (R) clk
17 |
18 | Capture          Launch
19 | Clock Edge:+     1000         0
20 | Drv Adjust:+       0         0
21 | Src Latency:+     0         0
22 | Net Latency:+     0 (I)      0 (I)
23 | Arrival:=        1000        0
24 |
25 | Setup:-          35
26 | Required Time:=   965
27 | Launch Clock:-    0
28 | Input Delay:-     500
29 | Data Path:-       422
30 | Slack:=          43
31 |
32 Exceptions/Constraints:
33 | input_delay       500              Q1_sdc.sdc_line_3_6_1

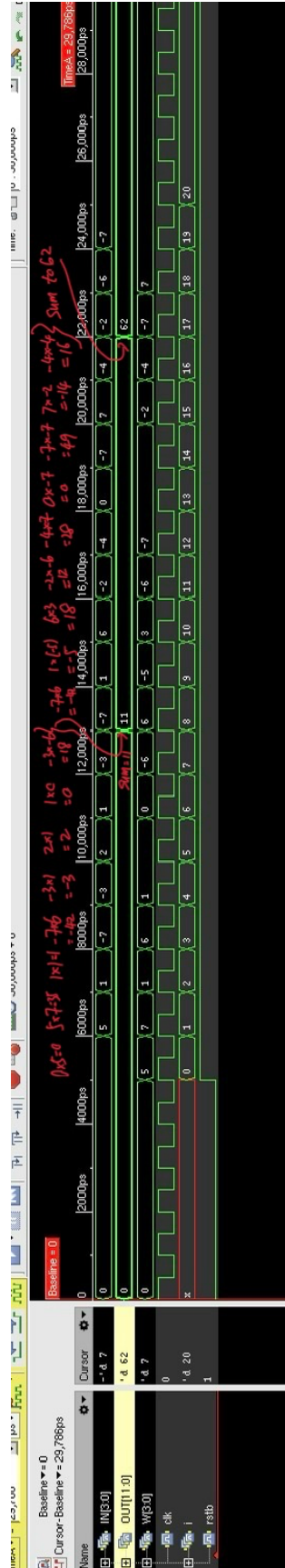
```

```

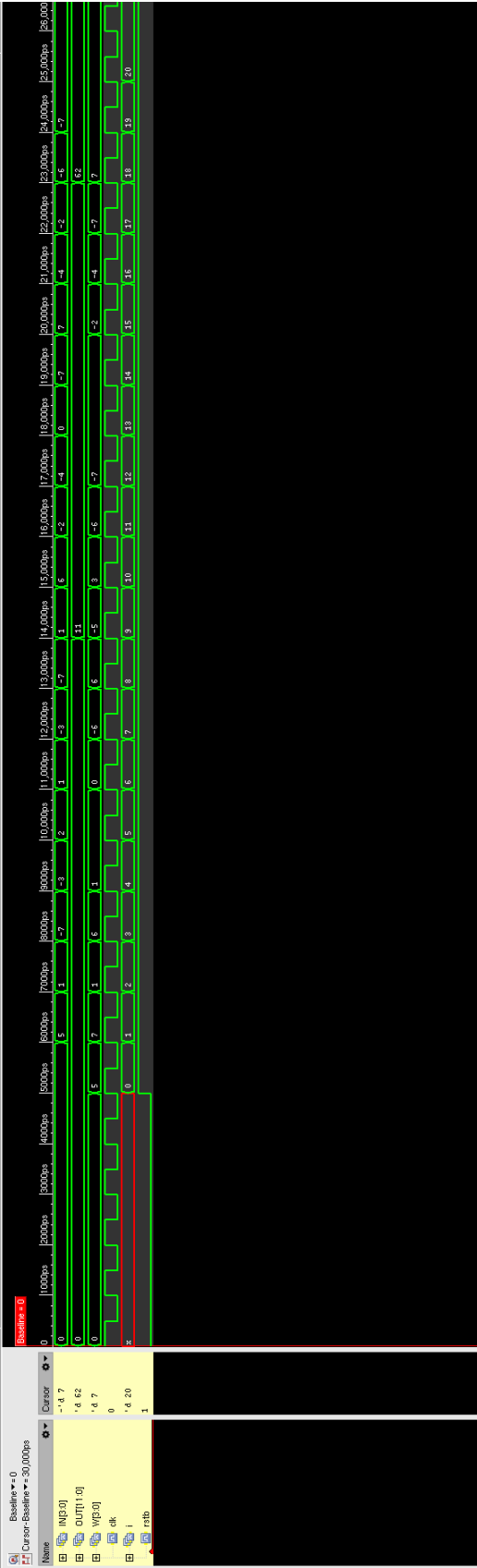
32 Exceptions/Constraints:
33 | input_delay          500          Q1_sdc.sdc_line_3_6_1
34
35 #-----
36 # Timing Point      Flags   Arc   Edge   Cell      Fanout Load Trans Delay
37 # Arrival Instance                                     (fF) (ps) (ps)
38 # (ps) Location
39 #-----
39 | W[1]              -      -      R      (arrival)    4  5.4    0
40 | 0          500    (-,-)
41 | g2324__7344/ZN    -      A2->ZN R      AND2_X1      1  3.5    12
42 | 33          533    (-,-)
43 | g2310__1474/S     -      A->S   R      HA_X1        1  3.1    28
44 | 55          588    (-,-)
45 | g2297__6083/S     -      CI->S   F      FA_X1        1  3.0    16
46 | 99          687    (-,-)
47 | g2280__2391/C0    -      CI->C0 F      FA_X1        1  3.0    15
48 | 73          760    (-,-)
49 | g2278__1309/C0    -      CI->C0 F      FA_X1        2  4.1    17
50 | 75          835    (-,-)
51 | g2274__2683/ZN    -      B1->ZN R      OAI21_X1     2  4.4    31
52 | 44          878    (-,-)
53 | g2268__3772/ZN    -      A->ZN   R      XNOR2_X1     1  1.4    17
54 | 43          922    (-,-)
55 | mult_reg_reg[6]/D <<< -      R      DFFR_X1      1  -      -
56 | 0          922    (-,-)
57 #-----
58 #-----

```

waveform (RTL):



waveform (synthesized):



Waveform. Notice that the clock time is 1ns

